Thu, Oct 8 at 8:14 PM

You said:

clarify what I did wrong here

x, y, z i.i.d, and uniform [0,1]

what is the probability of P(X<=YZ)

My thinking is YZ is a -log(yz)

so I think x as an axis, yz is another axis, so I form this 1 by 1 square on the first quadrant. and I draw out the boundary where pdf(x) = pdf (yz) and I calculate the area where pdf(x) is smaller than pdf(yz)?

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ChatGPT said:

Your intuition of **turning the probability into an area in a two-dimensional plane** is good! The mistake is that you're confusing **the value of a random variable** with **its probability density function (PDF)**.

There are actually two separate issues in your reasoning.

1. The distribution of YZ is not $-\log(YZ)$

Let

$$W = YZ.$$

Because $Y, Z \sim U(0,1)$ independently, the **PDF of** W is

$$\boxed{f_W(w) = -\ln(w), \quad 0 < w < 1.}$$

Notice the distinction:

- $W = YZ$ is a random variable.
- $f_W(w) = -\ln(w)$ describes the *density* of that random variable.

So you cannot replace YZ with $-\ln(YZ)$.

Also, since $X \sim U(0,1)$,

$$f_X(x) = 1, \quad 0 < x < 1.$$

2. Your square is valid, but the boundary is incorrect

You proposed drawing a unit square, with X on one axis and YZ on the other.

This is completely valid! We can use X and $W = YZ$ as our two axes.

However, your mistake is drawing the boundary where

$$f_X(x) = f_W(w).$$

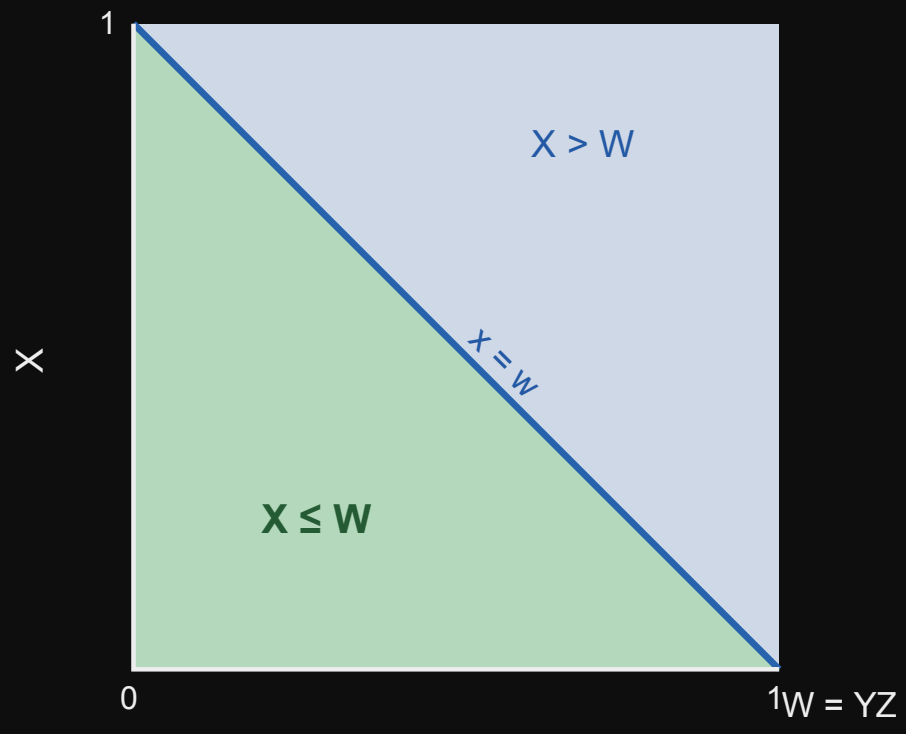
That would mean

$$1 = -\ln(w),$$

which has nothing to do with our event $X \leq W$.

Instead, the correct boundary is

$$\boxed{x = w.}$$



Correct event region in the unit square. Horizontal axis: W = YZ; vertical axis: X.

But there is one more important problem:

The probability is NOT simply the geometric area of the green triangle!

Why?

Because although X is uniform, $W = YZ$ is **not uniformly distributed**.

In fact, the joint PDF is

$$f_{X,W}(x,w) = f_X(x) f_W(w) = -\ln(w).$$

So probability must be calculated by integrating the joint density over the green region:

$$P(X \le YZ) = \int_0^1 \int_0^w -\ln(w)\, dx\, dw.$$

Integrating with respect to x:

$$= \int_0^1 -w \ln(w)\, dw$$

$$\boxed{P(X \le YZ) = \frac{1}{4}.}$$

3. There is an even simpler method!

Since $X \sim U(0,1)$, for any fixed $y, z \in [0,1]$,

$$P(X \le yz \mid Y = y, Z = z) = yz.$$

Therefore, by the law of total expectation,

$$\begin{aligned} P(X \le YZ) &= E[P(X \le YZ \mid Y, Z)] \\ &= E[YZ] \\ &= E[Y]E[Z] \\ &= \frac{1}{2} \cdot \frac{1}{2} \\ &= \boxed{\frac{1}{4}}. \end{aligned}$$

The key conceptual lesson

There are three different objects you should distinguish:

Object	Meaning
$W = YZ$	Value of a random variable
$f_W(w) = -\ln(w)$	Density of that random variable
$P(X \le W)$	Probability that one random variable is smaller than another

When you draw a region representing $X \le W$, the boundary always comes from **comparing the random variable values**, not their PDFs.

Your area method would give the answer directly if X and W were both independent uniform random variables, since the joint density would be constant at 1.

A question to test your understanding: Suppose X and W are independent, but $X \sim U(0,1)$ and W has PDF $f_W(w) = 2w$ on $[0,1]$.

Would $P(X \le W)$ still equal $1/2$, even though the favorable region is exactly half of the unit square? Why or why not?

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You said:

I think your previous boundary might be wrong? I think the boundary should be from lower left to top right?

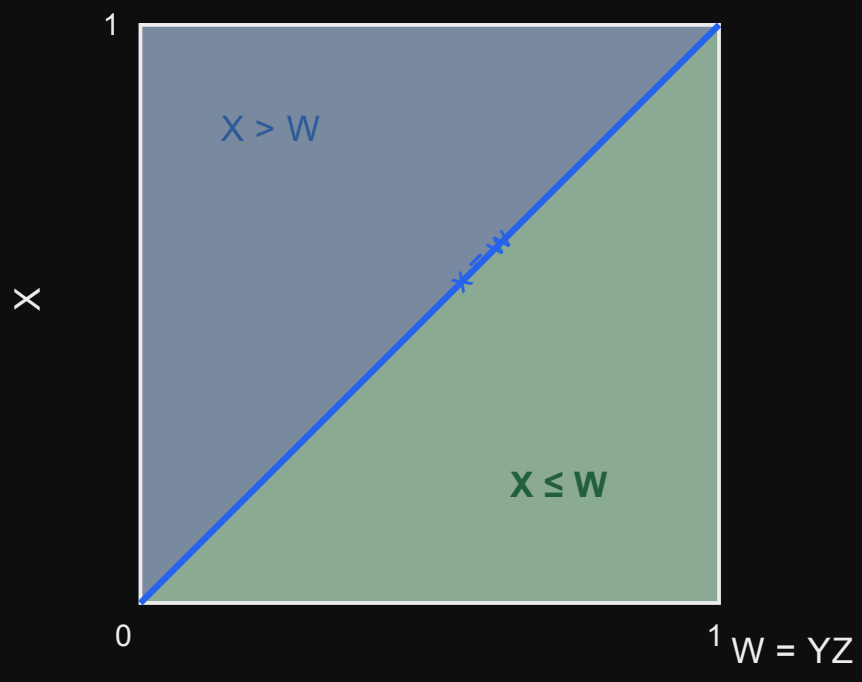
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ChatGPT said:

Yes, you're correct! **The boundary $x = w$ should run from the bottom-left to the top-right** in the usual Cartesian coordinate system.

My previous diagram was incorrect because I drew the diagonal from the top-left to the bottom-right.

Here's the corrected diagram:



With $W = YZ$ on the horizontal axis and X on the vertical axis:

- **Below the diagonal:** $X \leq W$, the event we want.
- **Above the diagonal:** $X > W$.
- **On the diagonal:** $X = W$.

The important point from the original explanation remains: the green triangle occupies half the square geometrically, but its probability is $1/4$, not $1/2$, because $W = YZ$ is not uniformly distributed.

Thank you for catching the diagram error!

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You said:

A question to test your understanding: Suppose \(X\) and \(W\) are independent, but \(X\sim U(0,1)\) and \(W\) has PDF \(f_W(w)=2w\) on \([0,1]\).
Would \(P(X\leq W)\) still equal \(1/2\), even though the favorable region is exactly half of the unit square? Why or why not?

For this question, I did not compute the integral (but I know how to). the result kinds of like a cubic but slice it by half as 长方体的一半（底面积是1*1的正方形）(which is the total region of the joint distribution of Pxw = 2w. and the area we are interested is a 棱锥, which is 1/3 of 长方体. 总的1是1/2个长方体，所以答案是2/3？

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ChatGPT said:

Yes! Your answer of $\boxed{\frac{2}{3}}$ is correct! And your geometric reasoning is a very interesting way to solve the problem without integration.

You've essentially turned a double integral into a **3D volume problem**.

There is just one geometric distinction I want to clarify.

1. Visualizing your idea

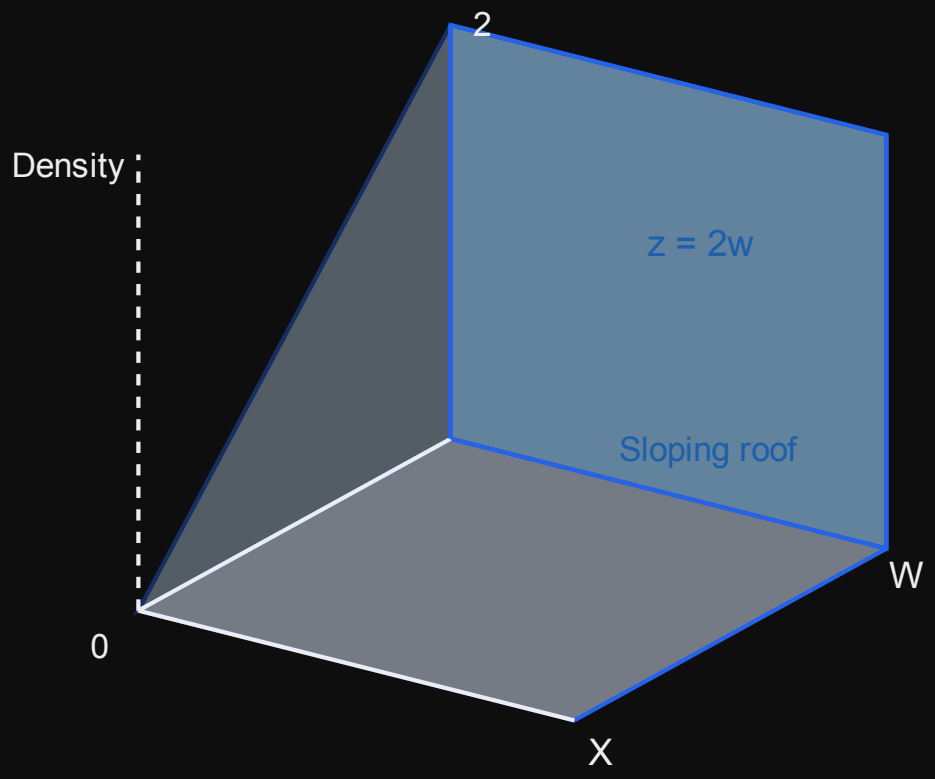
We have

$$f_{X,W}(x,w) = 2w, \qquad 0 \le x, w \le 1.$$

Imagine a three-dimensional graph with:

- x-axis: X
- w-axis: W
- z-axis: joint density $f_{X,W}(x,w) = 2w$

Because the density increases linearly with w, the solid under the PDF is a **triangular prism**（三棱柱）, not a pyramid.



Schematic of the full density solid: a triangular prism with a sloping roof $z = 2w$.
Its total volume is 1.

The full solid has volume

$$V_{\text{total}} = \frac{1}{2}(1)(2)(1) = 1.$$

Exactly as you suggested, this is half of a rectangular box whose dimensions are $1 \times 1 \times 2$.

2. The part corresponding to $X \leq W$

The event $X \leq W$ cuts the base square along its diagonal.

The excluded part, $X > W$, forms a **pyramid-like solid with volume** $1/3$. More precisely, it is a tetrahedron（四面体）.

The full solid has volume 1, so

$$P(X \leq W) = 1 - V_{\text{excluded}} = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}.$$

Alternatively, the full rectangular box has volume 2, and the excluded tetrahedron occupies $1/6$ of that box, giving volume $1/3$.

3. Why your intuition works

Your geometric solution captures an important statistical principle:

$$\boxed{\text{Probability} = \text{Volume under the joint PDF over the event region.}}$$

Even though the event $X \leq W$ takes up exactly half the square in the xw-plane, that half lies toward larger w, where $f_{X,W}(x,w) = 2w$ is larger.

So it contains **two-thirds of the probability mass**, rather than half.

One subtle correction to your description: the region we *exclude*, $X > W$, is the tetrahedron of volume $1/3$. The region we want, $X \leq W$, is the remainder of the triangular prism, with volume $2/3$.

This is a very useful geometric way to understand joint continuous distributions, especially when the PDF is linear.

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You said:

please also show me the ingetral way

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ChatGPT said:

Let's solve the same problem using a **double integral**, and connect each step to your geometric interpretation.

We know:

$$X \sim U(0,1), \qquad f_W(w) = 2w, \quad 0 \leq w \leq 1$$

and X, W are independent.

We want to calculate

$$P(X \leq W).$$

Step 1: Find the joint PDF

Because X and W are independent,

$$f_{X,W}(x,w) = f_X(x) f_W(w).$$

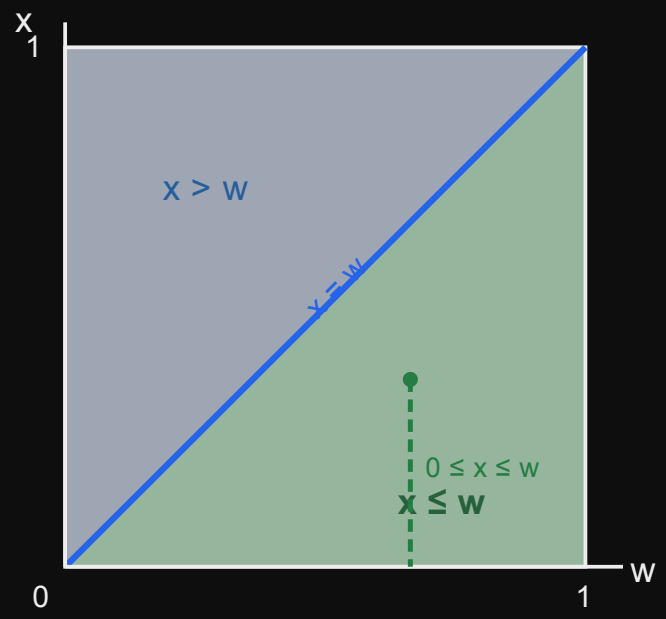
Since $f_X(x) = 1$,

$$\boxed{f_{X,W}(x,w) = 2w}$$

for $0 \le x, w \le 1$.

Step 2: Determine the integration region

The event $X \le W$ corresponds to the triangular region below the line $x = w$, when w is horizontal and x is vertical.



For any fixed w:

- w ranges from 0 to 1.
- x ranges from 0 to w.

Therefore,

$$\boxed{P(X \le W) = \int_0^1 \int_0^w 2w \, dx \, dw}$$

Step 3: Evaluate the double integral

First, integrate with respect to x, treating w as a constant:

$$\begin{aligned} P(X \le W) &= \int_0^1 \left[\int_0^w 2w \, dx \right] dw \\ &= \int_0^1 \left[2wx \right]_{x=0}^{x=w} \, dw \\ &= \int_0^1 2w^2 \, dw. \end{aligned}$$

Now integrate with respect to w:

$$\begin{aligned} P(X \le W) &= \left[\frac{2w^3}{3} \right]_0^1 \\ &= \boxed{\frac{2}{3}}. \end{aligned}$$

Step 4: What if we reverse the integration order?

We can also integrate with respect to w first.

For any fixed x, the condition $x \le w$ means w ranges from x to 1.

Thus,

$$\begin{aligned} P(X \le W) &= \int_0^1 \int_x^1 2w \, dw \, dx \\ &= \int_0^1 \left[w^2 \right]_{w=x}^{w=1} \, dx \\ &= \int_0^1 (1 - x^2) \, dx \\ &= \left[x - \frac{x^3}{3} \right]_0^1 \\ &= \boxed{\frac{2}{3}}. \end{aligned}$$

Step 5: Connection to your geometric reasoning

Notice that the inner integral in Step 3 produces

$$\int_0^w 2w\,dx = 2w^2.$$

Geometrically:

- The width of the slice is w.
- The height of the PDF is $2w$.
- The cross-sectional area is $w(2w) = 2w^2$.

The outer integral sums these cross-sectional areas from $w = 0$ to $w = 1$, giving the total volume $2/3$.

One more interesting connection: You can solve this entire problem with a single expectation:

$$\begin{aligned} P(X \le W) &= E[P(X \le W \mid W)] \\ &= E[W] \\ &= \int_0^1 w(2w)\,dw \\ &= \boxed{\frac{2}{3}}. \end{aligned}$$

This is the same result obtained through both the 3D geometric argument and double integration.

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